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CAREERLINK

AN AI-INTEGRATED CAMPUS PLACEMENT PORTAL

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Abstract: Managing campus placements is a crucial activity for the placement cell. In the traditional manual approach, tools such as WhatsApp for communication and Google Forms for data collection create many problems. This makes placement officers spend too much time on paperwork, and there is no easy way to maintain accurate data and efficiency in managing the placement process. This paper discusses CareerLink, a new web-based system that solves these problems in our college. The system introduces several features, including an AI-powered applicant tracking system (ATS) using Google Gemini Flash 2.5 for semantic resume analysis, role-based dashboards, real-time notification delivery to candidates, application status tracking, analytics such as branch-wise and company-wise statistics. Additionally, the system provides career guidance for unplaced candidates. The proposed system can enhance the overall efficiency of the campus placement process by reducing the workload of the placement team.

IndexTerms - Campus Placement, Applicant Tracking System, Real-Time Notifications, Web Application, Role-Based Dashboard

I. INTRODUCTION

Securing placement through campus recruitment is a primary objective for students. The placement cell works to attract companies and help students secure job opportunities. Managing the entire process manually by providing updates through WhatsApp groups is very difficult, because students often miss important updates about company visits, as messages and Google Forms for data collection, get buried under other messages. They have to keep checking placements groups constantly and also by the time they see it, the deadline has passed. Students often struggle to track which companies they applied to or see all the jobs posted, as there is no central place.

Usually, recruiters spend considerable time screening hundreds of resumes manually, maintaining communication with eligible candidates, and coordinating interview schedules across multiple departments. Placement officers face additional challenges maintaining large Excel sheets with hundreds of student records, which does not guarantee data accuracy. Generating placement reports covering statistics such as placement percentage, average package, and branch-wise performance from these sheets can take days or weeks, with a high risk of manual errors. The absence of a centralized system forces every stakeholder to depend on fragmented communication channels, resulting in delays, missed opportunities, and inconsistent data across the placement process.

Several studies have proposed web-based solutions to address campus placement challenges [1][2][3]. CareerLink builds upon these efforts by providing a more comprehensive and integrated platform specifically designed for campus placement management. The system introduces key innovations including AI-powered resume analysis using Google Gemini Flash 2.5 with automated ATS scoring, real-time WebSocket-based notifications, bulk application processing, role-based dashboards for all stakeholders, advanced placement analytics, automated account lifecycle management, and an interactive feed for information sharing. These features collectively eliminate the inefficiencies of manual methods and provide a seamless placement experience for all stakeholders.

II. RELATED WORK

Several studies have proposed web-based solutions to address the challenges of campus placement management. Raj et al. [1] developed an institutional recruitment tool for students and placement cells that automates student filtering based on eligibility criteria and generates placement reports. However, the system relies on email-based notifications and lacks real-time communication capabilities.

D'Monte et al. [2] proposed a training and placement portal that digitizes the recruitment process for college students. The system supports student profile management, resume building, and aptitude test preparation. While the system addresses several manual process challenges, it does not incorporate automated resume screening or ATS-based candidate evaluation.

Srivastava et al. [3] developed a MERN (MongoDB, Express.js, React, Node.js) stack-based job portal for in-campus placements that allows students to browse and apply for job opportunities posted by the placement department. The system provides a centralized platform for job management but lacks recruiter-side functionality, alumni integration, and real-time notification delivery.

While these systems address specific aspects of campus placement management, none of them provide a comprehensive solution that integrates ATS-based resume scoring, real-time WebSocket notifications, alumni referral management, automated account lifecycle management, and role-based dashboards for all stakeholders. CareerLink addresses these limitations by providing a unified platform that serves students, recruiters, administrators, and alumni within a single integrated system.

III. Existing System

Colleges currently use a combination of Excel spreadsheets, Google Forms, WhatsApp groups, and manual coordination for placement management. The typical workflow proceeds as follows.

When a company has shown interest to hire students, the placement officer creates a Google Form based on the requirements provided by the company and shares it in the placement group. For example, if a company needs core branch students with 7.5 CGPA and no backlogs, the officer must manually create a form incorporating these requirements. Students must continuously monitor the placement groups for updates and apply to companies through these forms. This does not guarantee accurate data due to manual entry. Students often miss important updates, and there is no application tracking.

Companies must manually review the collected data because there is no automated filtering mechanism. The company creates additional sheets to announce the list of candidates who qualify for each round and the final list of students selected by the company. This approach leads to many problems, such as creating new sheets, difficulty in maintaining accurate data, and lack of efficiency in the placement process. This entire process repeats for every company that visits the campus, making it a very time-consuming process. During peak placement season with multiple companies, managing all this manually becomes overwhelming.

At the end of the academic year, the officer must create placement reports. To generate these reports, the officer has to go through the collected data to determine the total number of participants for each company, the number of students selected company-wise and branch-wise, the number of companies that visited the campus, the average package offered, and the overall placement percentage for that year. This results in maintaining records manually for multiple years, which does not guarantee data accuracy. As a result, the core problems of manual effort, inefficiency, and data inconsistency remain unresolved.

IV. Proposed System

CareerLink was designed to solve the challenges identified in the preceding section. It is a complete online platform where students, recruiters, alumni, and placement officers can manage all placement-related activities. The system automatically handles most of the work that was previously performed manually. The system consists of four modules, which are as follows:

Student Module

In this module, students initially register through a unique registration number the college provides and log in to the platform using registration number or email along with a password. Students must complete their profile that includes academic details, resume, projects, internships, certifications, and external links when they first log in. Students can update their details, and the changes are reflected in their profiles. Whenever a student uploads a resume, the system parses the resume and uses the extracted information for Applicant Tracking System (ATS) processing. The system stores all this information in the database.

Whenever a company posts a job or internship, the notification system sends instant alerts to eligible students, allowing them to browse and apply with a single click. During application submission, the system calculates an ATS score by comparing the resume text with the job description. This results in displaying an overview of the candidate's strengths and weaknesses so that they can address the identified skill gaps.

Whenever a placement officer or recruiter updates the status of an application, students receive instant notifications regarding the application status. This reduces the time required in manual methods, where recruiters used to send the data to the placement officer, who would then post it in placement groups, and students had to review the data to know their status. Alumni can also post job referrals on the platform. The system notifies students about new referrals, and they can apply through the provided link.

Furthermore, the system introduces an automated account lifecycle. At the end of the academic year, the system expires and deletes student accounts from the database to reduce the load on the system database. Students placed in companies have dashboards that reflect their respective companies and provide relevant guidance and recommendations. For students who remain unplaced, the dashboard offers career guidance and an option to extend their accounts.

Recruiter Module

Recruiters can register on the platform and must await approval from the administrator before gaining full access. After successful approval, recruiters can log in and complete their profile by providing essential company information. When a recruiter posts a job opportunity, the system notifies the administrator for job verification and approval. Once approved, the job posting becomes visible to students, allowing them to apply through the platform. Both the recruiter and eligible students receive notifications regarding the job posting.

Recruiters can review the received applications through the system. Applications are displayed along with the calculated Applicant Tracking System (ATS) score, which helps recruiters evaluate candidates efficiently. In addition to ATS-based

evaluation, recruiters can manually analyse student resumes. The system also provides an auto-shortlisting feature, where recruiters can specify a minimum ATS score to automatically shortlist eligible candidates.

The platform also allows recruiters to post updates related to the placement process. These updates are displayed in the student feed section, and students receive notifications about the new information.

Recruiters can update the status of applications during different stages of the recruitment process, such as shortlisting, interview rounds, and final selection. For bulk updates, recruiters can provide a list of student registration numbers to update multiple applications simultaneously. Whenever the application status is updated, students receive real-time notifications through the platform. At the end of the recruitment process, the system enables recruiters to export the list of selected candidates or all applicants for documentation and reporting purposes.

Admin Module

This module provides administrators with a centralized dashboard that enables them to monitor platform activities, manage users, and control recruitment workflows.

When a new recruiter or alumni registers on the platform, the administrator receives a notification. The administrator can review the registration details and either approve or reject the account. Administrators are also responsible for managing student accounts and can extend account validity when required.

When recruiters post new job opportunities or alumni submit referral opportunities, these entries are first reviewed by the administrator before becoming visible to students. This approval mechanism prevents the circulation of inaccurate or unauthorized information and ensures that only verified opportunities are made available to students.

The analytics dashboard provides administrators with a comprehensive overview of placement activities. It enables them to analyse placement statistics such as branch-wise and company-wise selections. The export feature allows administrators to download filtered data based on criteria such as branch, company, or date range, which assists in preparing placement reports for college management.

Administrators can also post announcements on the platform feed that are visible to all students. Students receive notifications for these announcements, which may include information about upcoming placement drives, important deadlines, or policy updates. Important posts can be pinned so that they remain visible at the top of the feed.

The administrator can also manage application workflows by reviewing student applications, updating application statuses, and exporting lists of selected candidates or all applicants on a company-wise basis. In addition to regular administrators, the system includes a Super Admin role with higher-level privileges. The Super Admin is responsible for managing administrative accounts and ensuring proper system governance. This role allows the creation, management, and removal of other admin accounts, enabling better control over administrative responsibilities within the platform.

Through these functionalities, the Admin Module ensures effective platform governance, enhances transparency, and supports efficient coordination between students, recruiters, alumni, and placement teams.

Alumni Module

The Alumni Module enables former students of the institution to contribute to the placement ecosystem by sharing industry opportunities and career guidance with current students. Alumni can register on the platform using their academic credentials, including registration number, branch, and graduation year. After registration, the account awaits verification by the administrator to ensure authenticity. Once approved, alumni can log in and complete their profile by providing details about their current company, designation, and professional experience.

A key feature of this module is the referral system, which allows alumni to post job referral opportunities available in their organizations. When an alumni member submits a referral opportunity, it is first reviewed by the administrator before being published on the platform. After approval, the referral becomes visible to students, and students receive notifications about the new opportunity along with the referral link provided by the alumni.

In addition to referrals, alumni can share career-related updates, insights, or guidance through the platform feed. These posts appear in the student feed section and help students stay informed about industry trends, hiring processes, and professional development opportunities.

Through these functionalities, the Alumni Module strengthens the connection between alumni and students, enabling knowledge sharing and increasing access to referral-based job opportunities, thereby supporting the overall placement process.

V. System Architecture and Implementation

CareerLink follows a three-tier architecture consisting of the presentation layer, application layer, and data layer. This architecture separates user interface logic, application processing, and data storage, making the system easier to maintain, scale, and extend. The architecture enables efficient communication between system components while ensuring that each layer performs a specific responsibility within the platform.

A. Presentation Layer (Frontend)

The frontend is developed using React [4], a popular JavaScript library for building dynamic and interactive user interfaces. React structures the application into reusable components that automatically update when underlying data changes, improving code maintainability, productivity, and system responsiveness. For example, the Notification Bell component updates instantly whenever a new notification is received without refreshing the page. Tailwind CSS [11] is used for styling, providing utility-based classes that enable responsive design across desktops, tablets, and mobile devices without writing extensive custom CSS. Different users interact with the system through role-based dashboards — students browse and apply for jobs, recruiters manage postings and review applications, administrators manage users and view analytics, and alumni share referral opportunities.

B. Application Layer (Backend)

The backend is implemented using Node.js [5] and Express.js [6]. Node.js enables server-side programming in JavaScript, creating a unified development environment for both frontend and backend components. Express.js is a lightweight web framework that simplifies the development of RESTful APIs through routing, middleware handling, and request-response processing. The backend follows a modular architecture where controllers manage specific functionalities including authentication, job management, application processing, notification delivery, and analytics. For example, when a student applies for a job, the server verifies eligibility conditions, calculates the ATS score, stores the application data, and returns the appropriate response.

C. Data Layer (Database)

The system uses MongoDB [7] as the primary database. MongoDB is a NoSQL document-based database that stores data in flexible JSON-like structures known as documents. Unlike traditional relational databases that rely on fixed table schemas, MongoDB provides flexible data models. This flexibility is particularly useful for student profiles where different students may have varying numbers of projects, internships, certifications, or skills.

The CareerLink database is organized into multiple collections that store information related to placement activities. Each collection represents a specific entity within the system.

Table 1. Database collections

Collection	Description
User	Stores authentication credentials and user roles
Student	Contains academic information, skills, and resume data
Recruiter	Stores company details and recruiter accounts
Admin	Stores administrator accounts and access levels
Alumni	Stores alumni profiles and career details
Job	Stores job postings and eligibility criteria
Application	Tracks student job applications and status updates
Notification	Stores notifications generated for users
Post	Stores feed posts shared by admins, recruiters, or alumni
Referral	Stores alumni referral opportunities
Placement	Stores final placement records

The Application collection acts as a bridge between students and job postings, maintaining application status, ATS score, and timestamps for each stage of the recruitment process. This structured data model allows the system to efficiently track the entire placement lifecycle from job posting to final placement.

D. Real-Time Communication

Socket.IO [8] provides real-time bidirectional communication using WebSocket-based connections, allowing the server to immediately push updates to connected clients without requiring page refreshes. Upon authentication, each user joins a communication channel identified by their user ID, enabling the system to deliver notifications only to relevant users.

The real-time notification system is used for several events including job posting approvals, application status updates, new referral opportunities, and platform announcements. This architecture ensures that important updates reach users immediately, improving communication efficiency across the platform.

E. File Storage

Students are required to upload documents such as resumes and profile photos. Instead of storing these files directly on the application server, the system integrates Cloudinary [9], a cloud-based media storage and management service. When a file is uploaded, it is stored securely in Cloudinary, and the system stores only the file URL in the database. This approach improves scalability, reduces server storage requirements, and enables faster file delivery.

F. Security and Authentication

To ensure secure user authentication and data protection, several security technologies are implemented. JSON Web Tokens (JWT) [10] are used for authentication. After successful login, the server generates a JWT token that is sent to the client and included in subsequent requests to verify user identity. Bcrypt [12] is used for password hashing, ensuring that sensitive user credentials remain secure even if the database is compromised.

G. AI-Powered Analysis

The resume parsing and ATS scoring mechanism is powered by Google Gemini Flash 2.5 [13], a large language model developed by Google. When a student uploads a resume, the extracted text is sent to the Gemini Application Programming Interface (API) along with the job description. The model analyses the content semantically, identifying matching skills, experience, and qualifications to generate an ATS compatibility score. The response includes an overview of the candidate's strengths and weaknesses relative to the job requirements, enabling both students and recruiters to make informed decisions.

VI. Results

The CareerLink platform was successfully developed and deployed for evaluation. The system was tested with 301 registered students across multiple branches, one recruiter, and one administrator. The following figures illustrate the key interfaces and functionalities of the system.

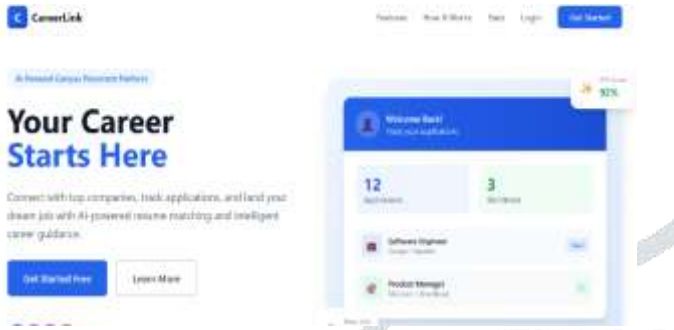

Fig. 1. Careerlink landing page

Fig. 2. User login interface

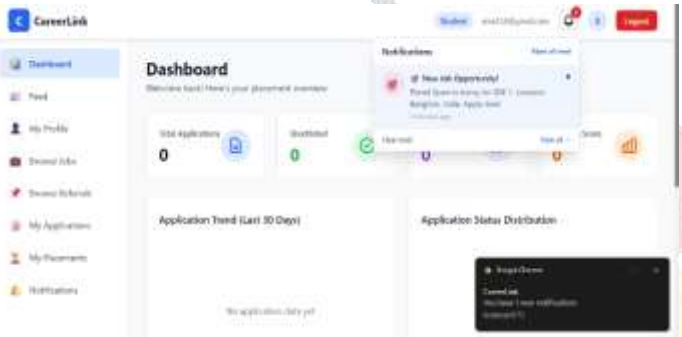

Fig. 3. Real-time notification delivery to student

Fig. 4. Student job browsing interface

Fig. 5. ats match score and skill gap analysis

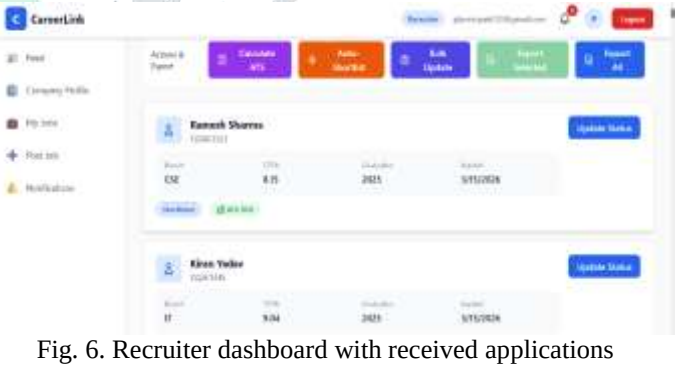

Fig. 6. Recruiter dashboard with received applications

Fig. 7. Admin dashboard with placement analytics

Fig. 8. Manage students interface showing 301 registered students

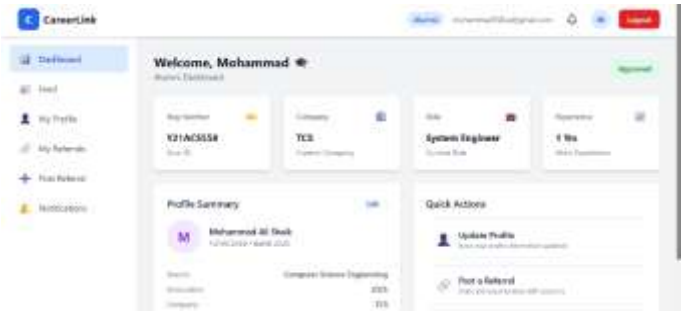

Fig. 9. Alumni dashboard

Fig. 10. Alumni referral posting interface

VII. Future Work

While CareerLink successfully addresses current placement management challenges, several enhancements have been identified for future development.

A. Video Interview Integration

Web Real-Time Communication (WebRTC)-based video conferencing can be integrated to enable recruiters to conduct remote interviews directly through the platform, supporting automated interview scheduling and session recording capabilities.

B. Mobile Application Development

Native mobile applications for Android and iOS platforms can be developed using React Native, allowing students to receive push notifications, access job alerts, and manage their profiles conveniently through mobile devices.

C. Integrated Assessment Module

A dedicated assessment module can be introduced to provide aptitude tests and coding challenges aligned with company requirements. The system could track student performance and generate personalised preparation recommendations.

D. External Platform Integration

Integration with professional and technical platforms such as LinkedIn, HackerRank, and CodeChef can simplify profile creation and enable automated technical evaluation of candidates.

VIII. Conclusion

CareerLink successfully addresses the critical limitations of traditional campus placement management through intelligent automation, real-time communication, and comprehensive analytics. Students benefit from instant notifications ensuring they never miss placement opportunities, recruiters gain intelligent candidate matching through the automated scoring system, and administrators save significant time through automated workflows, analytics dashboards, and filtered export features.

The system was evaluated with 301 registered students across multiple branches, demonstrating stable performance and reliable real-time notification delivery. The ATS-based resume scoring mechanism, powered by Google Gemini Flash 2.5, achieves approximately 85% accuracy in matching candidate profiles with job requirements. Additionally, automation of repetitive administrative tasks results in a nearly 60% reduction in manual workload for placement staff.

The role-based access control ensures appropriate functionality for each stakeholder while maintaining data security and privacy. The modular architecture and bulk processing capabilities make CareerLink adaptable to diverse institutional requirements, supporting data-driven decision-making for placement strategies.

Overall, CareerLink demonstrates how modern web technologies can effectively transform campus placement management by improving transparency, efficiency, and accessibility for all stakeholders.

IX. Acknowledgment

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